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29 December 1987
SUPERSEDING
MIL-S-80093A
15 July 1970

MILITARY SPECIFICATION

SHEARING MACHINES, ROTARY, RING AND CIRCLE, METAL, POWER OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, power operated, rotary shearing machines for ring and circle shearing, straight shearing, and for flanging and joggling, with equipment and accessories specified herein.

1.2 Classification. The shearing machines shall be of the following types. The sizes shall be as listed in tables I, II, or III. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Machines for shearing.

Type II - Machines for shearing and flanging.

Type III - Machines for shearing, flanging, and joggling.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

CFR 29, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951- Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street., Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators.

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of shearing, flanging, and joggling, as applicable, in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International

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System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, Chapter XVII, Part 1910 that are applicable to the machine itself. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements, should be specified in detail (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operation, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation .

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3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

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3.4 Components. The machine shall consist of not less than a frame, circle arm, cutter heads, shear cutters, flanging rolls and attachments, joggling rolls and attachments, straight shearing attachment, electrical equipment, and electrical controls.

3.4.1 Frame. The frame shall be of a stress-free construction of either iron or steel. The frame shall be the "C" style with an overhang, column, and base free to form a throat or worksheet opening. The base of the frame shall have provisions for securing the machine to a flat mounting surface.

3.4.2 Circle arm. Unless otherwise specified (see 6.2.1), the machine shall be provided with a clamp type circle arm. The circle arm shall be provided with clamping discs or shoes to hold and pivot worksheet for ring and circle shearing or circle flanging, as applicable for the type machine supplied. The circle arm shall be provided with means for laterally adjusting the discs or shoes to obtain any desired diameter within the applicable circle shearing and circle flanging ranges specified herein. The clamping discs or shoes shall be adjustable on machined bearing surfaces. The circle arm shall be easily removed or converted to permit unobstructed straight shearing.

3.4.3 Cutter heads. The upper and lower cutter heads shall be directly supported by the machine frame. The cutter heads shall be easily adaptable for the attachment of shear cutters for shearing and forming holddown rolls for flanging or joggling, as applicable for the machine type specified. The upper and lower cutter heads shall be provided with steel spindle shafts mounted on either antifriction ball or roller bearings, on bronze or ferrous metal bushings. Either the upper or lower, or both, spindle shafts shall be power driven. Means shall be provided for laterally adjusting either cutter,or both the upper or lower heads, in order to maintain alignment between the shear cutters or the forming rolls. The upper head shall be vertically adjustable in machined ways and shall be provided with means for locking the cutter head in a selected position. Unless otherwise specified (see 6.2.1), the upper cutter head for machine sizes with worksheet thickness shearing capacities greater than 1/4 inch shall be provided with a power downfeed for feeding the shear cutters or flanging rolls to penetrate or form the worksheet as the cut or flange progresses.

3.4.4 Shear cutters. The machine shall be provided with upper and lower shear cutters as normally furnished by the manufacturer for performing ring and circle and straight shearing as specified herein. The shear cutters shall be of tool steel, hardened to at least 55 on the Rockwell "C" scale. When required, special shear cutters shall be furnished as specified (see 6.2.1).

3.4.5 Flanging rolls and attachments. The type II and type III machines shall be provided with upper and lower flanging rolls and attachments normally furnished by the manufacturer for performing flanging as specified herein. The flanging rolls shall be of tool steel, hardened and ground for accuracy. When required, special flanging rolls and attachments shall be furnished as specified (see 6.2.1).

3.4.6 Joggling rolls and attachments. The type III machine shall be provided with joggling rolls and attachments normally furnished by the manufacturer for performing joggling as specified herein. The joggling rolls shall

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be tool steel, hardened and ground for accuracy. When required, special jogging rolls and attachments shall be furnished as specified (see 6.2.1).

3.4.7 Straight shearing (slitting) attachment. When specified (see 6.2.1), the machine shall be provided with a straight shearing (slitting) attachment for performing straight shearing as specified herein. The attachment shall include an adjustable straight edge for guiding worksheets past the shear cutters for straight shearing. The straight edge shall be adjustable along the throat of the machine frame on machined bearing surfaces. Means shall be provided for locking the straight edge in position. The straight shearing attachment shall guide the worksheets for shearing strip widths from not more than 2 inches wide up to widths not less than a measurement equal to 3/4 of the throat depth. The straight shearing attachment shall guide the worksheets for performing shear cuts straight within ±0.005 inch per linear foot of cut, parallel with a squared edge of worksheets.

3.4.8 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.9 Electrical system. The electrical system shall conform to NFPA 79. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.9.1 Primary input voltage. Unless otherwise specified (see 6.2.1), the machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phase, 60 Hz circuit. The machine shall be initially wired to operate from a 460 volt source.

3.4.9.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 50/60-Hz. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements for a drip proof enclosure. All motors shall meet the requirements of NEMA MG-1. Each motor shall have an identification plate to identify manufacturer, model, serial number, voltage, amperage, horsepower, phase, and frequency.

3.4.9.3 Control circuit voltage. The control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I, II, or III for the machine size specified.

3.6 Performance. The machine shall be capable of performance as specified in 3.6.1, 3.6.2, or 3.6.3 for the machine type specified.

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3.6.1 Type I machine. The type I machine shall be capable of shearing circles from square worksheets and rings from circular worksheets to comply with shearing capacity requirements of table I. The machine shall perform ring and circle shearing without ripping or wrinkling the worksheets. The machine shall not require a starting hole for making center cuts.

3.6.2 Type II machine. The type II machine shall be capable of shearing circles from square worksheets and rings from circular worksheets to comply with the shearing capacity requirements of table II. The machine shall perform ring and circle shearing without ripping or wrinkling the worksheets. The machine shall not require a starting hole for making center cuts. The Type II machine shall be capable of forming flanges from circular worksheets to comply with the flanging requirements of table II. The depth of the formed flanges shall be uniform throughout the circumference of the circle with no deviation more than 1/16 inch. Unless otherwise specified (see 6.2.1), the radius at the root of the flanges shall be as formed by the manufacturer's standard forming rolls.

3.6.3 Type III machine. The Type III machine shall be capable of shearing circles from square worksheets and rings from circular worksheets to comply with the shearing capacity requirements of table III. The machine shall perform ring and circle shearing without ripping or wrinkling the worksheets. The machine shall not require a starting hole for making center cuts. The type III machine shall be capable of forming flanges from circular worksheets to comply with the flanging requirements of table III. The depth of the formed flanges shall be uniform throughout the circumference of the circle with no deviation more than 1/16 inch. Unless otherwise specified (see 6.2.1), the radius at the root of the flanges shall be as formed by the manufacturer's standard forming rolls. The type III machine shall be capable of forming joggle offsets up to 1/2 inch deep. The dimensions of the offsets shall be uniform without a deviation of more than 1/16 inch per foot for any cross-sectional dimension.

3.6.4 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished as described.

3.8 Optional equipment. Optional equipment specified in the following subparagraphs shall be furnished with each machine, as specified. (see 6.2.1).

a. Air operated clamping device for circle arm.

b. Special circle arm for flanging bilged heads.

c. Power positioning for circle arm.

d. Other optional equipment or features required.

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TABLE I. Type I sizes and characteristics.

Characteristics	Sizes													
	1	1A	3	4	5	6	7	9	10	11	12	13	14	15
Circle cutting diameter, inches														
-Minimum, not more than	6	6	6	10	6	8	8	10	8	14	10	18	18	26
-Maximum, not less than	22	60	60	45	60	60	72	60	72	72	144	144	144	144
in														
Worksheet thickness, inches														
-Not less than	0.059	0.075	0.134	1/8	1/8	3/16	3/16	1/4	1/4	3/8	3/8	1/2	3/4	1
Cutting speed, feet per minute (FPM)														
-Not less than	22	22	43	60	22	10	10	20	20	12	12	10	10	10
Throat depth inches														
-Not less than	9	18	18	18	30	36	36	25	36	36	36	48	48	48
Motor horsepower														
-Not less than	1/2	3/4	2	1-1/2	1	2	2	5	3	5	5	5	15	20

NOTE: Shearing capacity shall be performed in mild steel of not less than 55,000 PSI.

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TABLE II. Type II sizes and characteristics.

Characteristics	Sizes								
	1	2	3	4	5	6	7	8	9
Circle cutting diameter, inches									
-Minimum, not more than	6	6	8	8	8	10	18	18	26
-Maxiumum, not less than	60	60	58	72	72	144	144	144	144
in									
Worksheet thickness, inches									
-Not less than	0.075	0.134	0.164	3/16	1/4	3/8	1/2	3/4	1
at a									
Cutting speed, feet per minute,									
-Not less than	22	22	43	10	20	12	10	10	10
Circle flanging diameters, inches,									
-Not more than	6	6	19	8	8	10	18	18	26
-Not less than	60	60	73	72	72	144	44	144	144
in									
Worksheet thickness, inches,									
-Not less than	0.060	0.075	0.164	1/8	3/16	5/16	3/8	1/2	5/8
To produce flange depth, inches,									
-Not less than	7/8	7/8	1-1/8	1-1/2	1-1/8	2	2-1/4	2-1/2	2-3/4
Throat depth, inches,									
-Not less than	13	15	17	36	36	36	48	48	48
Motor horsepower									
-Not less than	3/4	1-1/2	2	2	3	5	10	15	20

NOTE: Shearing and flanging capacity shall be performed in mild steel of not less than 55,000 PSI.

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TABLE III. Type III sizes and characteristics.

Characteristics	Sizes								
	1	2	4	5	6	7	8	10	12
Circle cutting diameter, inches,									
-Minimum, not more than	8	8	8	8	8	14	20	24	26
-Maximum, not less than	48	48	48	60	72	72	72	72	72
in									
Worksheet thickness, inches,									
-Not less than	0.074	0.134	1/8	3/16	1/4	3/8	1/2	3/4	1
at a									
Cutting speed, feet per minute,									
-Not less than	22	43	22	10	20	12	10	10	10
Circle flanging diameter, inches,									
-Not more than	8	8	8	8	8	14	10	24	26
-Not less than	48	48	48	60	72	72	72	72	72
in									
Worksheet thickness, inches,									
-Not less than	0.059	0.104	0.074	1/8	3/16	5/16	3/8	1/2	5/8
Flange depth inches,									
-Not less than	7/8	1-1/8	7/8	1-1/2	1-1/8	2	2-1/4	2-1/2	2-3/4
Joggling in workpiece thickness, inches,									
-Not less than	0.059	0.104	0.074	1/8	3/16	1/4	3/8	1/2	3/4
Throat depth inches,									
-Not less than	60	20	30	36	36	36	48	48	48
Motor horsepower									
-Not less than	3/4	1-1/2	1	2	3	5	10	15	20

NOTE: Shearing, flanging, and joggling capacity shall be performed in mild steel of not less than 55,000 PSI.

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3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication plate. A lubrication plate shall be securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test,or the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification, including compliance with all referenced standards.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 60 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

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4.7.2 Performance test. The machine shall be tested for performance by shearing a worksheet of mild steel with a tensile strength of not less than 55,000 PSI, worksheet thickness and capacity shall be as specified in table I, table II, or table III as applicable for the machine type specified. The instruments and methods used in performing the test shall be at the option of the contractor, provided they accurately determine the capacity of the machine.

4.7.2.1 Type I machine. The type I machine shall be operated to verify conformance with the shearing requirements of 3.6.1 and table I by performing rotary shearing operations as follows:

a. Shearing a circle of the smallest diameter specified for the machine.

b. Shearing a ring with the outside diameter equal to the largest diameter specified for the machine. The inside diameter of the ring shall not be more than four inches smaller than the outside diameter. Ring and circle shearing shall be performed without ripping or wrinkling the worksheets.

c. When a straight shearing (slitting) attachment is furnished (see 6.2.1), three straight shear cuts of at least 36 inches in length each shall be performed. The straight shear cuts shall be straight to within $\pm$ 0.0005 inch per foot of cut, parallel with a squared edge of the test piece guided by the straight edge of the straight shearing attachment.

4.7.2.2 Type II machine. The type II machine shall be operated to verify conformance with the shearing requirements of 3.6.2 and table II by following the same procedure specified in 4.7.2.1. The type II machine shall be operated to verify conformance with the flanging requirements of 3.6.2 and table II by performing flange forming operations as follows:

a. Forming a 1/2 inch deep circular flange on the smallest diameter specified for the machine.

b. Forming the deepest specified flange on the largest diameter specified for the machine.

Flange forming shall be performed to provide a uniform flange depth throughout the circumference of the circles without a deviation of more than 1/16 inch.

4.7.2.3 Type III machines. The type III machine shall be operated to verify conformance with the shearing and flanging requirements of 3.6.3 and table III by following the same procedure specified in 4.7.2.2. The type III machine shall be operated to verify conformance with the joggling requirements of 3.6.3 and table III by forming three joggle offsets of at least 1/2 inch deep and at least 36 inches in length each. The dimensions of the joggle joints shall remain uniform without a deviation of more than 1/16 inch per foot for any cross-sectional dimension.

4.7.3 Electromagnetic interference control tests. When specified, the machine shall be tested for conformance with 3.3.8. The test shall be accomplished in accordance with MIL-STD-462.

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4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are primarily intended for ring and circle shearing and flange forming operations that are frequently required in shipyard and aircraft maintenance shops.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size of machine required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction are required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and subceptibility required (see 3.3.8).

j. Circle arm, if different (see 3.4.2).

k. Power downfeed, if different (see 3.4.3).

l. Special shear cutters, if required, specify type (see 3.4.4).

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m. Special flanging rolls and attachments, if required specify type (see 3.4.5).

n. Special joggling rolls and attachments, if required, specify type (3.4.6).

o. Straight shearing (slitting) attachment, if required (see 3.4.7).

p. Primary input voltage, if different (see 3.4.9.1).

q. Motor enclosure, if different (see 3.4.9.2).

r. Size and capacity, if different (see 3.5).

s. Radius of root for flanges, if different (see 3.6.2 and 3.6.3).

t. Additional performance features, if required (see 3.6.4).

u. Standard equipment, if different (see 3.7).

v. Specify and fully describe optional equipment, if required (see 3.8).

w. Specify applicable repair parts, if required (see 3.9).

x. Hour meter, if not required (see 3.11).

y. First article inspection, if different (see 4.4).

z. Quality conformance inspection, if different (see 4.5).

aa. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Ring and circle shearing
Shearing machine
Straight shearing

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Air Force - 99

Review activity:
Air Force - 84

User activities:
Army - AV, AR

Preparing activity:
DLA-IP

(Project 3445-0108)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-80093B	Military Specification Shearing Machine, Rotary, Ring and Circle, Metal, Power Operated

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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